Exploratory signed-Frobenius rho calibration ladder

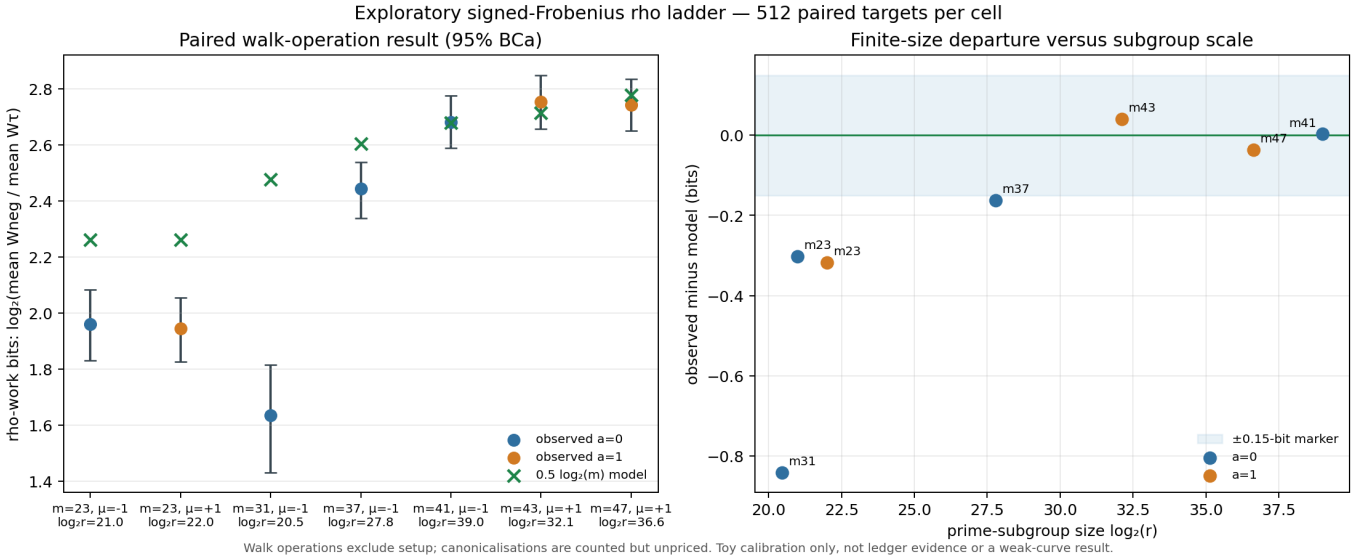
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Status: **complete exploratory observation; not ledger evidence or a measured bound.** This run calibrates source-curve rho counters on word-sized Koblitz groups. It neither executes an isogeny transfer nor tests a named NIST curve for weakness.

Outcome

All 10,752 planned executions recovered their planted logarithms, and the runner verified each returned scalar against its target. A subsequent full same-binary deterministic replay reproduced all 10,752 executions with identical counts and found zero problems. The primary analysis uses 512 paired targets per cell and 50,000 paired target-level BCa resamples.

The result is a scale transition, not a blanket confirmation. The three cells with `log2(r) >= 32` (`m=41,43,47`) measured 2.681, 2.754, and 2.741 rho-work bits beyond negation; their pointwise 95% intervals contain the random-map model `0.5 log2(m)`. All four smaller-subgroup cells undershot the model, especially `m=31`, whose degree is larger than `m=23` but whose selected prime subgroup is only about 20.46 bits. Field degree alone is therefore not a safe experimental scale variable.



Walk operations exclude setup; canonicalisations are counted but unpriced. Toy calibration only, not ledger evidence or a weak-curve result.

Primary walk result

`W` is `counters.walk_operations`: steps plus look-ahead additions plus cycle-escape doublings, excluding setup, walk starts, and verification. The estimand is `log2(mean(W_neg)/mean(W_tau))`; it is not the mean of per-target ratios. Cells are not pooled.

cell	log2(r)	cofactor	neg/tau walk ratio (95% BCa)	observed bits (95% BCa)	model bits	residual
a=0, m=23	20.999	4	3.890 [3.555, 4.237]	1.960 [1.830, 2.083]	2.262	-0.302
a=1, m=23	22.001	2	3.848 [3.546, 4.156]	1.944 [1.826, 2.055]	2.262	-0.318
a=0, m=31	20.457	1492	3.109 [2.698, 3.520]	1.636 [1.432, 1.815]	2.477	-0.841
a=0, m=37	27.781	596	5.437 [5.056, 5.806]	2.443 [2.338, 2.537]	2.605	-0.162
a=0, m=41	39.000	4	6.415 [6.014, 6.849]	2.681 [2.588, 2.776]	2.679	+0.003
a=1, m=43	32.113	1894	6.745 [6.307, 7.205]	2.754 [2.657, 2.849]	2.713	+0.041
a=1, m=47	36.636	1318	6.687 [6.277, 7.130]	2.741 [2.650, 2.834]	2.777	-0.036

The `+/-0.15`-bit band in the graph is a frozen descriptive marker, not a promotion rule. It contains the point residuals for `m=41,43,47`, not those for the four smaller-subgroup cells. No asymptotic exponent was fitted.

What "three to four rho bits" means

The baseline matters.

- Incremental signed-Frobenius work saving beyond an already standard negation quotient is modelled as `0.5 log2(m)` bits.
- Gross saving beyond unquotiented plain rho is `0.5 log2(2m)`, exactly one half-bit larger in the random-map model.

At the three larger-subgroup cells here, the observed incremental walk saving is 2.68--2.75 bits and the observed gross plain-to-signed saving is 3.18--3.24 bits. For the named NIST K curves, the structural model gives 3.67--4.58 bits beyond negation (12.77x--23.90x fewer expected transitions), or 4.17--5.08 bits beyond plain rho. Roughly four rho-work bits is operationally material, but it is not a loss of four subgroup-entropy bits, a new attack, a wall-clock gain, or evidence that an isogenous representative is weak.

Secondary accounting

cell	counted-GAE neg/tau	plain/tau walk bits	wall neg/tau (L0, descriptive)
a=0, m=23	3.033x	2.339	1.261x
a=1, m=23	2.652x	2.450	1.369x
a=0, m=31	2.530x	1.981	0.959x
a=0, m=37	4.888x	2.801	2.988x
a=0, m=41	6.393x	3.223	5.477x
a=1, m=43	6.522x	3.185	4.638x
a=1, m=47	6.635x	3.241	5.553x

The signed and negation arms count but do not price orbit canonicalisations, and they omit different numbers of them. Each affected counted-GAE total is incomplete, while their ratio has no automatic lower- or upper-bound direction. Wall timing came from an unisolated virtual host (`L0`), so even its BCa interval is descriptive rather than a performance result. The `m=31` walk advantage disappearing in wall time illustrates why the transition model cannot substitute for complete costing.

The run retained 589 local trail-cap restarts. These are charged cycle-control events inside verified overall solves, not timeouts or exhausted executions. Cycle, look-ahead, distinguished-point, useless-collision, setup, start, verification, and unpriced-canonicalisation totals remain in `analysis.json`.

Archived C-study setup reanalysis

The earlier C studies counted fixed target-dependent setup in both arms. The exact common term is `(QPOW - 1) + R_ADD`: 155 group operations at `m=37` and 167 at `m=41`. A new deterministic 50,000-resample reanalysis gives:

study	archived total-op ratio (95%)	setup-excluded target-dependent ratio (95%)	sqrt(m)	residual bits
m=37	5.728 [5.616, 5.840]	6.041 [5.920, 6.165]	6.083	-0.0098
m=41	6.361 [6.198, 6.528]	6.370 [6.205, 6.538]	6.403	-0.0076

This explains the apparent `m=37` shortfall in the old total-op comparison. The setup-excluded C count still contains walk starts, look-ahead work, cycle escapes, and abandoned work; it excludes recovery/verification and is not the same counter as this ladder's `walk_operations`. Those independent-arm studies use different implementations and are contextual replications only, never pooled with the fresh data. The source and output are `reanalyse_archived_setup.py` and `archived-setup-reanalysis.json`.

Design, validity, and integrity

The design was committed before launch in `PROTOCOL.md` and `spec.json`. Post-launch editorial clarifications are isolated in `PROTOCOL-CLARIFICATIONS.md`; they change no grid, sample count, seed, estimand, analysis, or stopping rule.

- producer: `aburan28/crypto` commit `0dc6b7f4a03253e7c7641f99700f7f423f8d07e0`, observed clean before build;
- dependency lock archive: `producer-Cargo.lock`, 29,115 bytes;
- lockfile SHA-256: `4ddc5ef76119493b41352c64a551fbd3fea3b5789644949ee775975b38150bca`;
- binary SHA-256: `f2164ce16c6ae17c408777dc2448e8f226a42ca8c4d3da1d00aeff7419ec1f88`;
- spec: `ECS1h37e1cc41d056`, file SHA-256 `751769ef2eeb6c9946f9822c14956c9c2617335859e46752ad7f1e510586c950`;
- session: `ECBS1h5a5461ff22d7`;
- uncompressed records SHA-256: `d82f6319a75e16d0185323b69e2e4b0760f2753510ec45f026b7004148a49a3c`;
- full replay/audit SHA-256: `74bfadbd5c6d3e0da09fc826dc5323365db5361577008d5ea5a31a7747b71cb7`.

The producer session sealed the binary digest but did not embed Git metadata or the dependency-lock digest; `build-provenance.json` records that boundary instead of pretending those fields were session-sealed. The raw records and expanded plan are preserved as `records.jsonl.zst` and `plan.json.zst`. The complete replay receipt is `audit.json`. `MANIFEST.sha256` seals every other file in this study directory.

Reproduce the analysis

From the `crypto-autoresearcher` repository root:

```
python3 research/koblitz_rho_ladder_20261013/analyze.py \
  --records research/koblitz_rho_ladder_20261013/records.jsonl.zst \
  --spec research/koblitz_rho_ladder_20261013/spec.json \
  --session research/koblitz_rho_ladder_20261013/session.json \
  --audit research/koblitz_rho_ladder_20261013/audit.json \
  --output research/koblitz_rho_ladder_20261013/analysis.json \
  --csv research/koblitz_rho_ladder_20261013/paired-results.csv
python3 research/koblitz_rho_ladder_20261013/reanalyse_archived_setup.py
MPLCONFIGDIR=/tmp/matplotlib-koblitz \
  python3 research/koblitz_rho_ladder_20261013/plot.py
```

The failed setup/launch attempts and successful commands are retained in `ATTEMPTS.md`, `run-command.txt`, and `verify-command.txt`.

What this changes—and what it does not

This run improves the source-action baseline in three ways: it adds seven same-harness cells, exposes the finite-subgroup-size effect, and separates walk work from setup. It does not discharge any map, dual, subgroup-transfer, endpoint-action, or one-time-transfer gate in the isogeny protocol. It says nothing new about conductor gaps, P/B curves, sect113r1, or the existence of a weak endpoint.

The next high-value measurement is fully priced canonicalisation and fixed-transition throughput, including larger prime subgroups and full-size NIST K fixed-length blocks. The next isogeny result still requires an explicit map/dual and the one-time transfer control; another nominal degree point cannot replace either obligation.